MILITARY SPECIFICATION

CARTRIDGE, ILLUMINATING, M301A3 WITH FUZE, TIME M84A1 FOR 81MM MORTARS LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of illuminating cartridge, designated as Cartridge, Illuminating, M301A3, w/Fuze, Time, M84A1

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-P-46568 - Primer, Percussion, M71A1E1, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810 - Environmental Test Methods
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for percent defective

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DRAWINGS

ORDNANCE CORPS

9241848	-	Container, Ammo, Fiber, PA43, for Cartridge, 81MM, Illum, M301A3 in Jungle Wrap
9241850	-	Protective Bag Assembly for Cartridge, 81MM, Illum, M301A3
9241849	-	Box, Packing, Ammunition for Cartridge, 81MM, Illum, In Jungle Wrap
9220705	-	Cartridge, Illuminating, 81MM, M301A3E1 with Fuze, Time, M84A1

PUBLICATIONS

ET-9220705	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable specifications and drawings.

3.2 Loaded cartridge.-The loaded cartridge shall comply with all requirements specified on Drawing (dwg.) 9220705 and with all requirements specified in applicable specifications.

3.3 Disassembly torque

3.3.1 Fuze.-The fuze shall withstand a disassembly torque of 200 inch pounds when tested as specified in 4.3.1.

3.3.2 Fin assembly.-The fin assembly shall withstand a disassembly torque of 100 inch pounds when tested as specified in 4.3.1.

3.4 Proving Ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.1.

3.4.1 Firing.-The cartridge shall not fail to fire.

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3.4.2 Fuze.-The fuze shall function and cause separation of the body tube assembly and the tail cone.

3.4.3 Functioning.-The projectile loading assembly shall function satisfactorily (see 6.4) and shall provide effective illumination (see 6.5) for 60 seconds, min.

3.4.4 Security of parts.-There shall be no metal parts failure in the bore of the weapon or in the first approximate 200 yards of horizontal range.

3.4.5 Horizontal range.-No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.4.6 Fuze Burning Time.-The fuze assemblies shall function and shall comply with the burning time in Table A when tested in accordance with 4.3.2.1.

TABLE A

Fuze Setting (sec)	Charge Increment	Burning Time (sec)
20	4	18.10 - 21.90
30	8	28.00 - 32.00

3.5 Workmanship

3.5.1 All parts and assemblies shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.5.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

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3.5.3 Surface finish.-The requirements for surface finish shall be as specified on the applicable drawings.

3.6 First Article Sample.-This specification makes provisions for first article ballistic inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for susbmission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded complete round cartridges shall contain:

a. Loaded projectiles from one interfix lot number from one manufacturer.

b. Loaded ignition cartridges from one interfix lot number from one manufacturer.

c. Loaded time fuzes from one interfix lot number from one manufacturer.

d. Loaded primers from one interfix lot number from one manufacturer.

e. Propellant increments made from not more than one lot of propellant from one manufacturer.

f. Fin assemblies from one interfix lot number from one manufacturer.

4.2.1.1 First Article Ballistic Testing

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4.2.1.2 Submission of ballistic assemblies.-The contractor shall submit a first article ballistic sample as designated by the contracting officer for determination of conformance to drawing and specification requirements. After the fuze and projectile first article samples have been accepted, the first article ballistic sample shall be tested and shall consist of the following:

a. Twenty-five complete rounds assembled with inert loaded illuminant loading assemblies (suggested filler: sand and laminac) and inert fuzes.

b. Forteen completely loaded complete rounds fuzed, painted and marked.

4.2.1.3 Metal parts security.-Twenty-five (25) of the cartridge metal parts shall be tested as shown in the phases in Table B. All of the cartridges shall be inert loaded to service weight and will be fired from a weapon for which the round is standard. The propellant charge will be adjusted to obtain approximately 8400 pounds per square inch. Inert fuzes shall be used on all of the metal parts security and recovery test rounds. Pressure data shall be taken for each round fired.

TABLE B

Phase	Mortar Elevation (deg)	Initial Production Sample
Security	45	10
Recovery	Optional	15

Recovery - Rounds fired for recovery which satisfy the requirements of 3.4.4 are to determine quality of ammunition produced. Examination shall be made after recovery for evidence of body or tail cone deformation, dishing or breakup of the projectile other than by impact. If the proving ground determines that deformation, dishing or breakup of the cartridge has resulted, the results of the test shall be referred.

4.2.1.4 Cartridge functioning.-After successful completion of the projectile and fuze specification first article tests as well as the cartridge metal part security test the cartridge functioning tests shall be performed. The fifteen loaded cartridges shall be inspected for conformance to any or all of the requirements of this specification and any or all of the requirements of the applicable drawings at the contractor's plant prior to submission to the proving ground for proving ground testing. Seven of the sample projectiles shall be fired at the Charge 4 zone while the remaining seven will be fired at the Charge 8 zone. Cartridge fuze function time shall be evaluated in accordance with MIL-STD-414, Section B, Table B3, AQL 4.0 percent.

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4.2.1.5 Recovery.-The proving ground shall make the necessary arrangements to recover all of the rounds fired for recovery during the metal parts security test and any or all malfunctioning parachute and illuminant assemblies which may occur during the cartridge functioning test. Arrangements for the recovery of malfunctions shall be for the first article samples only, unless directed by the procuring activity.

4.2.1.6 Rejection.-If any internal parts, inert or loaded; or external metal parts or assemblies fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. If during the metal part security or recovery test any cartridge fails to comply with the requirement of 3.4.4, the first article sample shall be rejected. In addition, if during the cartridge functioning test any loaded cartridge fails to comply with the requirements of 3.4.1, 3.4.2, 3.4.3, 3.4.4, 3.4.5 and 3.4.6, the first article cartridge functioning sample shall be rejected. The Government reserves the right to terminate its inspection and testing upon any failure in the first article sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a complete first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings, (see 6.6). Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Cartridge, prior to assembly of fuze, fin assembly and adapter (see dwg. 9220705).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.50 percent		
201.	Threads damaged	Visual	01001
202.	Pettman Cement missing where required	Visual	01002
203.	Evidence of poor workmanship, (see 3.5)	Visual	01003

4.2.2.2 Holder, increment, front (see dwg. 9220708 covering a detail of dwg. 9220705).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Diameter of hole, min	Gage	02001
102.	Thickness, min	Gage	02002
Minor:	AQL 1.0 percent		
201.	Evidence of poor workmanship, (see 3.5)	Visual	02003
202.	Finger broken or damaged	Visual	02004

4.2.2.3 Cartridge assembly (see dwg. 9220705)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Fuze not set safe	Visual	03001
2.	Safety pin improperly engaged	Visual	03002
3.	Primer missing or above flush	Visual	03003
4.	Propellant increment missing	Visual	03004
5.	Fin broken or cracked	Visual	03005
6.	Ignition cartridge missing or inverted or primer flash hole obstructed	Gage	03006
7.	Warning marking missing or illegible	Visual	03007

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Major:	100 Percent Inspection		
101.	Fin distorted	Visual/ Gage	03008
Major:	AQL 1.0 percent		
102.	Total length, max	Gage	03009
103.	Torque test failures of fuze, (see 4.3.1)	Test	03010
104.	Torque test failure of fin assembly (see 4.3.1)	Test	03011
105.	Any component damaged or incorrectly assembled (except as described in any other defect contained herein)	Visual	03012
106.	Thread sealing compound does not cover a minimum of one half the number of a male primer threads for a full 360 degrees (prior to assembly)	Visual	03013
107.	Sealing compound bead does not cover crimp on red end of ignition cartridge or mating surface on bottom of ignition cartridge cavity in housing (as applicable) for a full 360 degrees (prior to insertion of ignition cartridge into fin cavity)	Visual	03014
108.	Sealing compound bead does not cover crimp on yellow end of ignition cartridge or mating surface for a full 360 degrees (prior to insertion of ignition cartridge into fin cavity)	Visual	03015
Minor:	AQL 1.50 percent		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	03016
202.	Marking misleading or unidentifiable, including incorrect color code	Visual	03017
203.	Evidence of poor workmanship, (see 3.5)	Visual	03018

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4.2.2.4 Unsealed Fiber Container (see dwg. 9241848 and 9241850)

Categories	Defects	Method of Inspection	Code No.
Major:	AQL 1.0 percent		
101.	Neck ring missing	Visual	04001
102.	Support tube missing	Visual	04002
103.	Dessicant not secure to cartridge	Visual	04003
104.	Protective bag assembly not secure to cartridge	Visual	04004
105.	Twist tie wire not secure	Visual	04005
Minor:	AQL 2.5 percent		
201.	Bag cut or torn	Visual	04006
202.	Filler disc assembly missing	Visual	04007
203.	Cushion missing	Visual	04008
204.	Dessicant missing	Visual	04009
205.	Round cannot be removed by hand	Visual/ Manual	04010
206.	Evidence of poor workmanship, (see 3.5)	Visual	04011

4.2.2.5 Sealed fiber container wrapped in barrier material prior to wax jungle wrap (see dwg. 9241848 and dwg. 9230175).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Sealing tape incomplete or badly wrinkled	Visual	05001
102.	Barrier material improperly assembled	Visual	05002
103.	Barrier material improperly folded or taped	Visual	05003
Minor:	AQL 2.5 percent		
201.	Gap between body and cover of container exceeding 1/8 inch	Visual	05004
202.	Contents loose	Manual	05005
203.	Tear tab length inadequate	Visual	05006
204.	Marking misleading or unidentifiable	Visual	05007
205.	Marking missing or incomplete	Visual	05008
206.	Evidence of poor workmanship, (see 3.5)	Visual	05009

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4.2.2.6 Sealed Fiber container after waxing-Jungle Wrap (see dwg. 9241848 and dwg. 9230175).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wax coating incomplete or damaged	Visual	06001
Minor:	AQL 1.00 percent		
201.	Marking missing or unidentifiable	Visual	06002
202.	Kraft paper improperly placed on container	Visual	06003

4.2.2.7 Sealed wood packing box (see dwg. 9241849).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Board broken or split	Visual	07001
102.	Strapping missing, broken or loose	Visual	07002
103.	DOD symbol missing or incorrect	Visual	07003
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	07004
202.	Strapping improperly assembled	Visual	07005
203.	Marking misleading or unidentifiable	Visual	07006
204.	Carseal missing, unsealed or improperly positioned	Visual/ Manual	07007

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of Primer, Percussion, M71A1E1 (see Specification MIL-P-46568 (MU)). If the total elapsed time between original acceptance of the primer lot and the assembly of that lot into the loaded cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46568 (MU) immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the loaded cartridges (see 6.8).

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4.2.3.2 Profile and alignment (see dwg. 9220705) - Major defect - Code No. 03019.-The shell shall be gaged 100 percent for profile and alignment. Any shell which fails to comply with the requirement specified shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4. The gaging shall be done just prior to installing the propellant.

4.2.3.3 Proving Ground

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified in Table I, sixty sample cartridges shall be selected for this test. The cartridges shall be observed for compliance with the requirements specified. Testing shall be in accordance with 4.3.2.1.

4.2.3.3.2 After three consecutive lots have complied with the criteria of 4.2.3.3.1, fifty sample cartridges shall be selected for this test. The cartridges shall be observed for compliance with Table I and the requirements specified. Testing shall be in accordance with 4.3.2.1. In the event that three successive lots fail to meet the requirements of 4.2.3.3.2, the sampling provisions of 4.2.3.3.1 shall again apply. When three consecutive lots have again successfully passed the requirements contained therein, the provisions of 4.2.3.3.2 shall be reinstated.

TABLE I

Category	Defect	Code No.	Sample Size (60 or 50) Paragraphs 4.2.3.3.1 and 4.2.3.3.2		
			ACC	REJ	NOTE 1
Critical	Premature burst or metal parts separation	03020	0	1	
Major	Failure to fire	03021	1	2	
Major	Fuze functioning	03022	2	3	
Major	Fuze functioning time	03023	SEE NOTE 2		
Major	Horizontal range less than 80% of average range	03024	1	2	
Major	Projectile does not ignite or provides for effective illumination for 40 seconds or less	03025	4	5	Note 3
Minor	Projectile provides effective illumination for more than 40 seconds but less than 60 seconds	03026	7	8	Note 3

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Note 1: The lot shall also be rejected if in a sample of 60 or 50 rounds a total of seven or more Major defects is encountered.

Note 2: The lot shall also be rejected if the percent defective for fuze ballistic burning time as computed with the use of MIL-STD-414, Section B, Table B3, AQL 4.0 percent, exceeds the maximum allowable percent defective for each setting.

Note 3: If the fuze performance of any round in the sample exceeds the timing requirements specified, and the round functions properly in all respects with the exception that the "effective illumination" time is less than sixty (60) seconds because the illuminant and parachute assembly hit the ground while still burning, defect code 03021 or 03022 shall not be additionally assessed against that round.

4.2.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-9220705 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Torque test.-The disassembly torque shall be applied to the applicable component while holding the adjacent mating component in such a manner as to avoid possible loosening or damage to any component of the assembly.

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4.3.2 Proving ground.-The test shall be performed at a Government Proving Ground in accordance with the applicable Test Procedure. All of the cartridges shall be fired from an 81MM Mortar for which the round is standard.

4.3.2.1 Ballistic tests.-One half of the sample cartridges shall be fired with an ignition cartridge and four propellant increments (charge zone four) and one half of the sample cartridges shall be fired with an ignition cartridge and eight propellant increments (charge zone eight), and in accordance with elevation specified in Table II. Observation shall be made for compliance with the requirements specified for security of parts, functioning, functioning time, horizontal range and "effective illumination" (see 6.5). All flight times shall be determined by infra-red chronograph. Fuze functioning time shall be reported to at least the nearest 0.01 sec, illuminant burning time to the nearest 0.5 second and horizontal range on the ground to a point under the position of fuze functioning to the nearest 5 meters measured by optical triangulation. The cartridges shall be conditioned at 75 plus or minus 5 degrees Fahrenheit for a minimum of 4 hours prior to testing and fired at this temperature. Any malfunction occurring during the test shall be reported in the official firing record. Meteorological data shall also be reported in the official firing record. Muzzle velocity shall also be recorded and reported for each round fired.

Charge Zone	Fuze Setting	Mortar Elevation (degrees)
4	20 sec	60
8	30 sec	60

4.3.2.1.1 Test Validity.-If for any reason the proving ground considers the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with a new sample.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

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5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with drawings 9241848 and 9241850. In addition, the opened container (cover removed from body) shall be dried in a circulating oven for a minimum of 24 hours at a temperature of 120°F ± 5°F prior to packing. After temperature conditioning, the container shall be closed tightly for shipment, storage, etc. until no more than one-half hour prior to final packing of the cartridge in the container. The cartridge must be packed, and the fiber container sealed (Jungle Packed) within 24 hours of the time the container was removed from the oven. If the fiber containers are to be left on the assembly line or if for some reason the packaging in the fiber containers is to be stopped for more than one-half hour, the containers are to be closed tightly until they are ready to be used. Storage or transportation in excess of 24 hours after removal from conditioning oven shall require the cartons be packaged in moisture proof bags. Internal loading parts consisting of felt padding and chip board filler shall be included for moisture control by same oven drying requirements and subsequent packaging in moisture proof bags if necessary.

5.2 Packing

5.2.1 Level A.-The packaged cartridges shall be packed in accordance with dwg. 9241849.

5.3 Marking

5.3.1 The marking of the packages and pack shall be in accordance with dwg. 9241848 and dwg. 9241849.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army design, mandatory for use

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Satisfactory functioning.-Satisfactory functioning shall be defined as ejection of the parachute and illuminant assembly and proper suspension and ignition of the illuminant such as to assure "effective illumination".

6.5 Effective illumination.-Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

6.6 It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.7 Statements of findings shall show the specific analytical results of each test or inspection performed as required by the applicable drawing or specification together with the specified acceptable limits.

6.8 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.9 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2 and DA4, Dover, New Jersey 07801.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1315-A-597

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

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